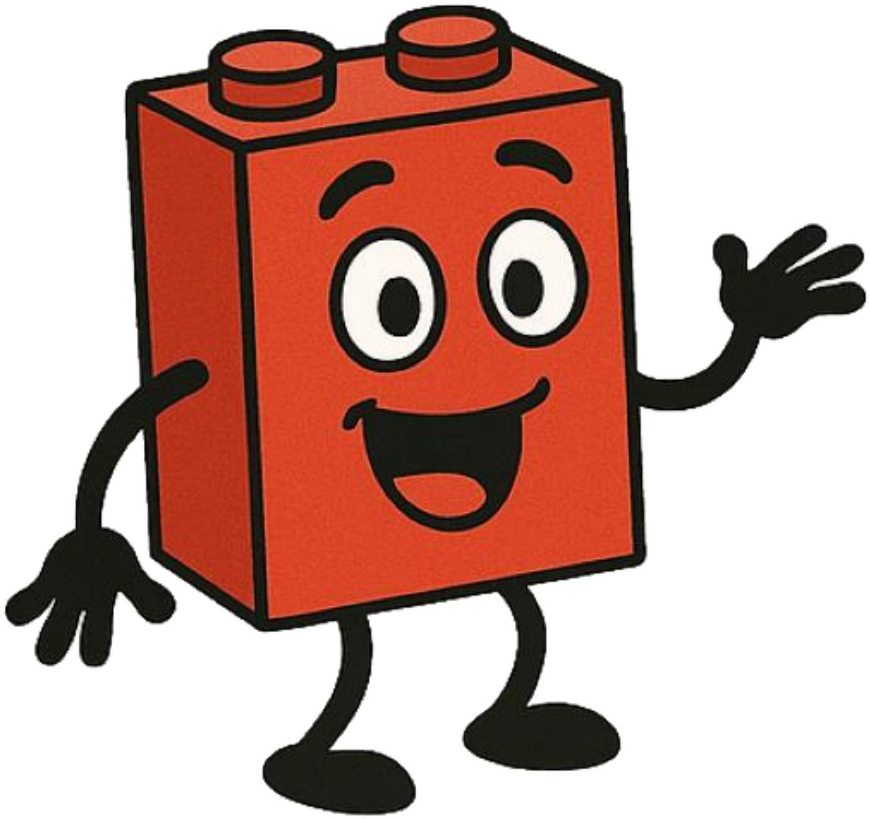
AUDIOBRIX



USER MANUAL

universität wien

Master's Internship P2 – Merlin Waldhör
Supervisor: Dr. Oliver Hödl

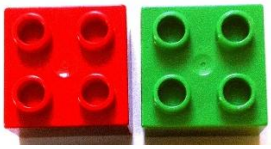
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Arduino Nano Every



Miscellaneous blocks



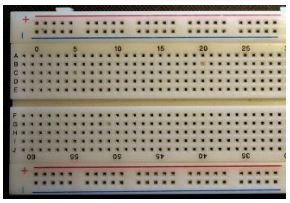
Aluminium Foil



Resistors



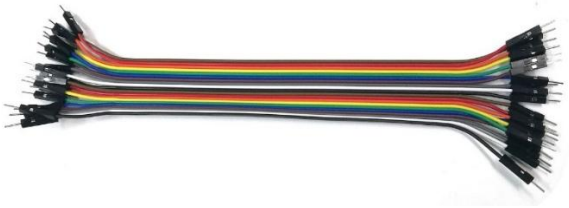
Conductive Tape
(Wide + Narrow)



Breadboards



USB Micro Cable



Jumper Cables

Inhalt	Anzahl
Base plate with wooden substructure	1
Copper Strip Wide	1
Copper Strip Narrow	1
Arduino Nano Every	1
2x2 Playing Bricks	> 7
2x2 Playing Bricks Drilled	8
2x4 Playing Bricks	> 25
Mini Breadboards	2
Screws 10mm M3	1
Micro USB Cable	1
Wire ~40cm	1
Resistors 1k Ohm	>7
Resistors 2k Ohm	>33
Aluminium Foil	1
Jumper Wires MM	20
Jumper Wires FM	20

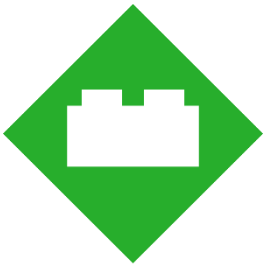
Make sure that all the specified individual parts are present at the beginning.

HELLO!

Welcome to the instructions of the AudioBrix kit! In this tutorial you will learn new things about electronics, computer science and music.

This manual will guide you through the setup, wiring and playing of the AudioBrix sequencer. At the end of this guide you will be able to compose your own music tracks and make music using Duplo blocks.

Please note that the AudioBrix kit is available in two versions. Make sure you use the instructions according to your version.



AudioBrix Relaxed Edition:
This kit is marked with a green symbol and contains already assembled parts.
All you have to do here is do the wiring.



AudioBrix Explorer Edition:
This kit is marked with a red symbol. In order to play the instrument, you must first assemble the game material.



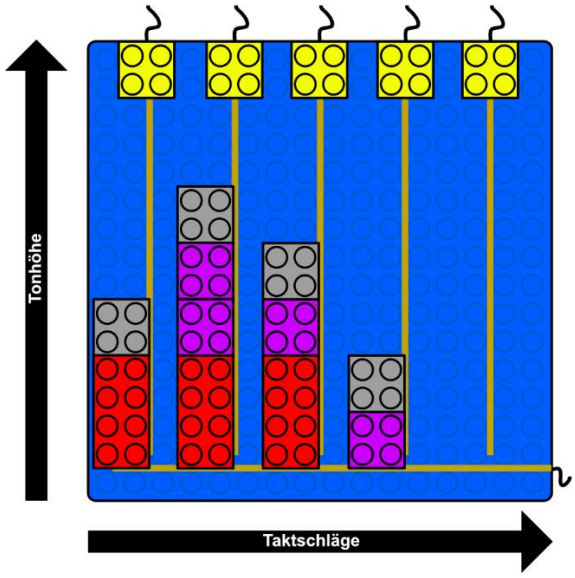
The Explorer Edition requires work tools.
Never work unsupervised.

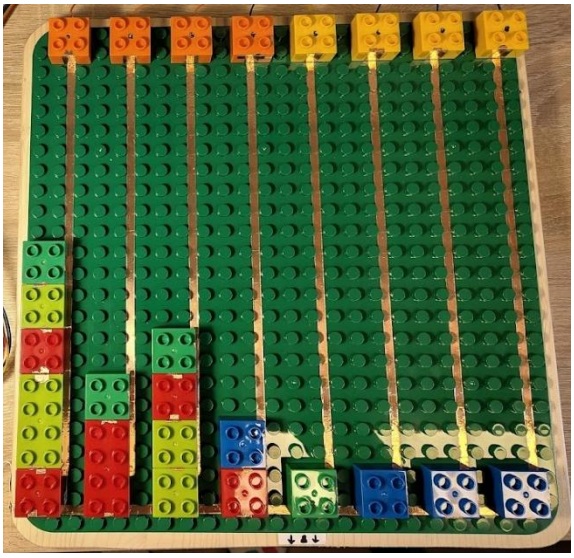
The AudioBrix figure was illustrated using AI.
This AudioBrix kit manual is for educational purposes only.
The English version was translated from German using Microsoft Translator.

LET'S BUILD MUSIC!

With the AudioBrix kit we create an instrument from Duplo blocks with which we can bring music to life.

Before we really get started, let's get a rough overview of the instrument. Here you can see two representations: A simplified visualization (left) and the real instrument (right).





The instrument works like a sequencer. A **sequencer** plays the notes from left to right in a loop.

Think of it like a sheet of music that is played repeatedly. Each note is represented by one or more building blocks per column.

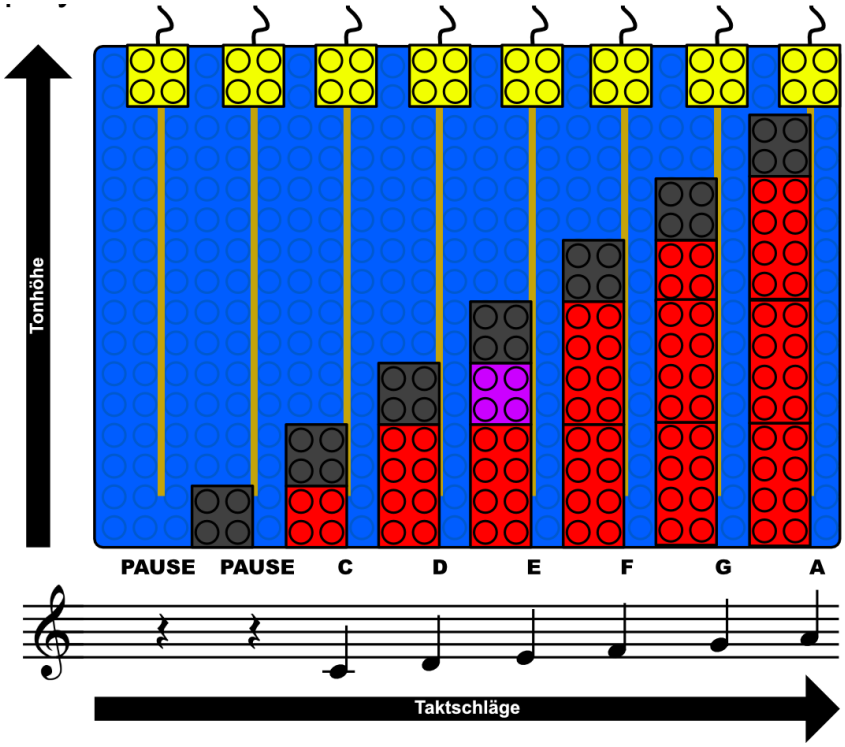
The more building blocks are placed in a column, the higher the grade. Fewer building blocks mean, for example, lower tones. ("Note height"). We will learn more about this later.

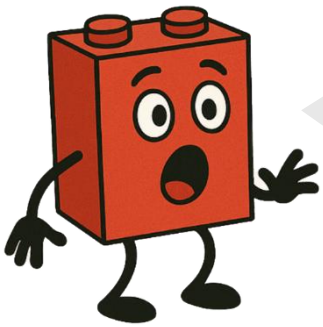
SEQUENCER DESIGN

In order for us to really play, we first need to understand how the instrument represents notes and beats.

We can easily transfer classic notes from music to the AudioBrix sequencer. In the image below you can see the display of notes on the AudioBrix instrument.

If no (note) blocks are placed in a column, this is considered a break.



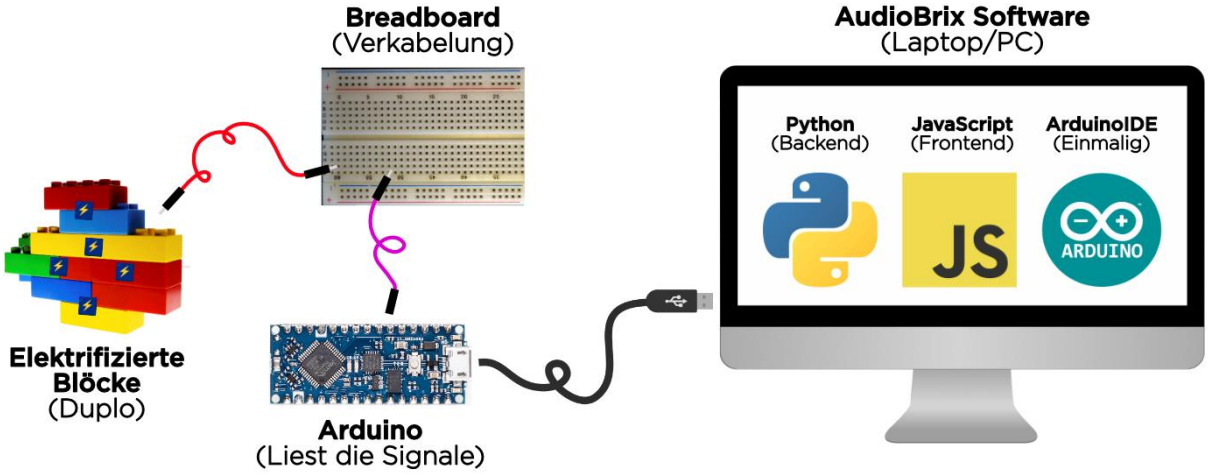


Plus One!?

To make a note audible, we need an additional block for each column (beat). This is shown here in dark gray. This block does not represent an additional note height – we will see why later in electrification!

WIE GEHT DAS?

Sehen wir uns kurz den technischen Aufbau an, bevor wir uns das Instrument weiter ansehen:



Electrified blocks:
This includes the modified blocks and base plate. The blocks close a circuit, which then creates measurable resistances.

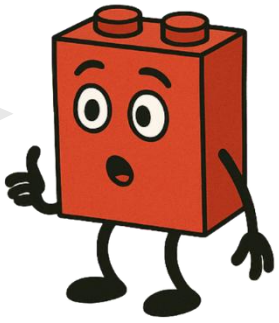
Breadboard:
This is where most of the wiring takes place.

Arduino:
The Arduino is a development board with sensors – it reads the signals from the blocks and sends these signals to the computer.

AudioBrix Software:
The AudioBrix software runs on the computer. This is based on the programming languages Python and JavaScript. It takes over communication with the Arduino, converts the signals into notes and takes care of playing the notes.

Online video tutorials

On dev.waldhoer.org/audiobrix there are various tutorials for assembling and helpful tricks - feel free to take a look if you get stuck somewhere!



ELECTRIFICATION

Enough reading – let's get started! The core of the instrument must first be built: the blocks and base plate!

In the next steps, we will prepare the base plate and make the blocks. After we have all the blocks ready, we will proceed with the wiring.

Have the following things ready for electrification:

- Aluminum foil
- Resistors
- Conductive Tape

In addition, you will need:

- Scissors
- Working document
- Hot glue gun
- Gloves
- Small ruler or triangle

Make enough space on your table so that you can work properly and don't lose anything.

Divide up your work materials and work steps if you work with others – this way you can finish faster!

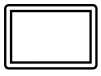
When doing manual work, keep the following information in mind:



Risk of injury! Use gloves when working with the hot glue gun.



Never work unsupervised! During the work, a supervisor must always be nearby.



Use a work pad and make sure your work surface is large enough.



Working with electricity: The Arduino works with 5V electricity. This is harmless for humans. For more information, see page 36.

TECHNICAL SETUP

To set up the instrument, we first need:

- **Base plate** with nine conductive tracks
- **Notepads** equipped with resistors
- **Bridge blocks** connecting note blocks to the railways
- **Sensor blocks** at the end of each lane

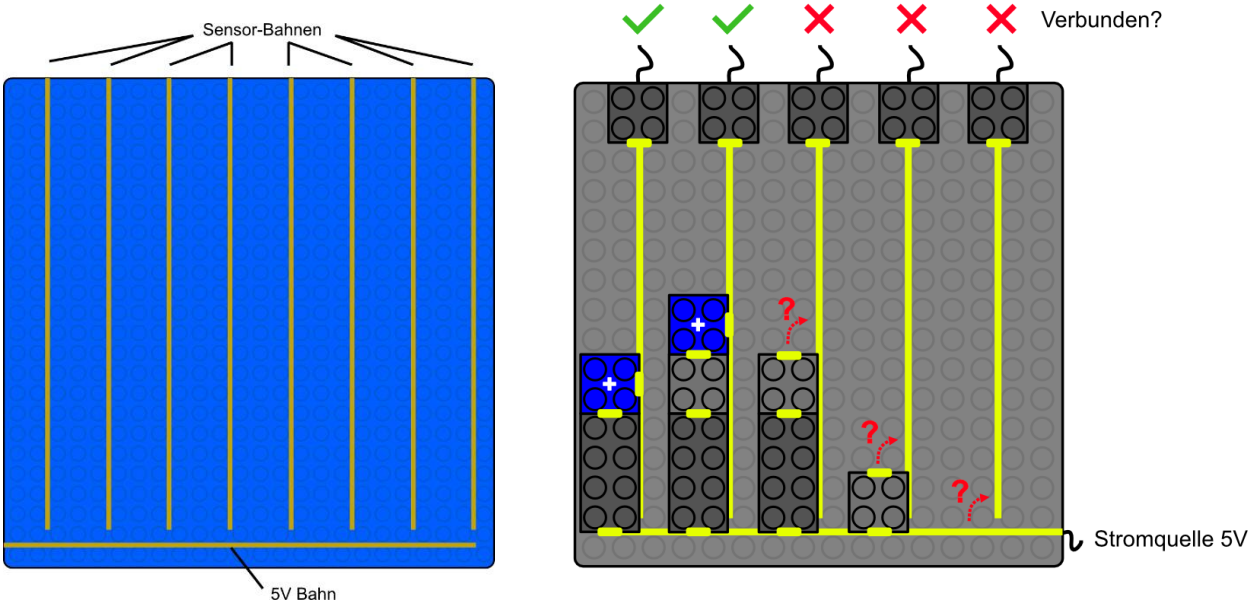
Let's take a look at the base plate. This has various conductive adhesive strips (we call these "**strips**").

- The lower, transverse track is the **power supply**. This track supplies all note blocks with 5 volts.
- The vertically running tracks are the **sensor paths**. They each run to an end piece (sensor block) that connects the sensor track to the cabling.

The sensor tracks are not directly connected to the power source. We want to **interpose resistors** (**in the form of note blocks**).

The note blocks are placed a pin to the left of the sensor tracks (so that the edges of the note blocks touch the track – not in the middle of the track). This is important, because otherwise the current would take the shortest path and skip the note blocks.

After the last note block (the uppermost one in each column), we therefore place a connecting block (bridge block). This is shown in blue in the picture on the right. Without a bridge block, the sensor is not connected.



BASE PLATE

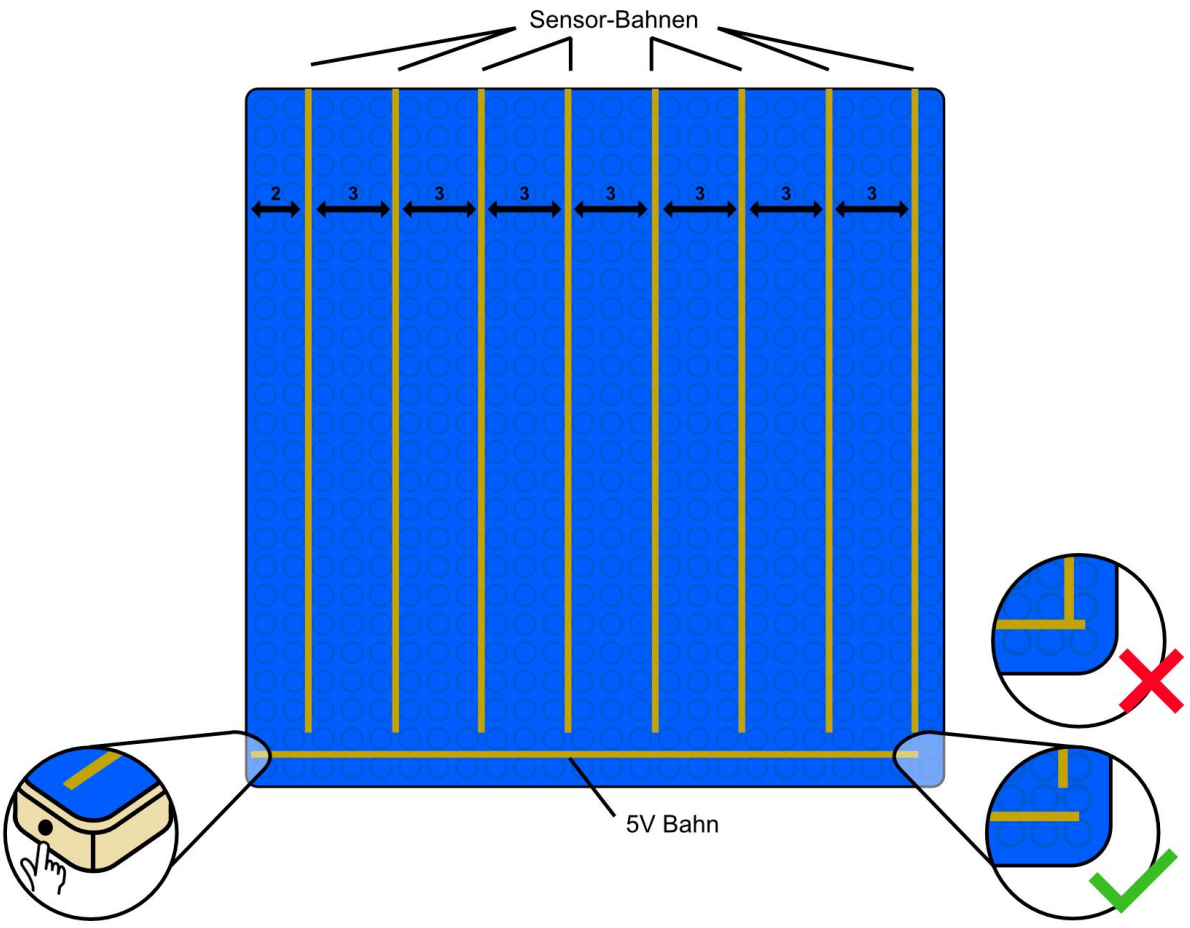
First of all, we need to prepare the base plate. This board must first be equipped with conductive tape.

In total, we need nine long **narrow** adhesive strips.

- There is a 5 volt track that runs vertically
- There are eight sensor orbits that run horizontally

Cut **off 9 long narrow adhesive strips** and glue them to the base plate as in the picture. Attach the strips so that the 5 volt strip is at the level of the pre-drilled hole.

Make sure the spacing is correct and no tape is touching.



Each adhesive strip should be applied carefully so that there are no tears or major scratches on the adhesive strip.

BLOCK OVERVIEW

The next step is to electrify the units.

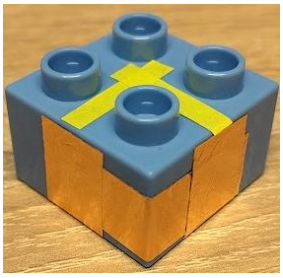
There are three types of blocks that we need to produce in different quantities. **Each block type has its own color**. The correct block colors are indicated in your toolkit.

- **8 x Bridge blocks**: see page 12
- **8 x Sensor Blocks**: see page 13
- **Notepads**: see page 15
  - ~8 x Small notepads (2x2)
  - ~25 x Large Note Pads (2x4)

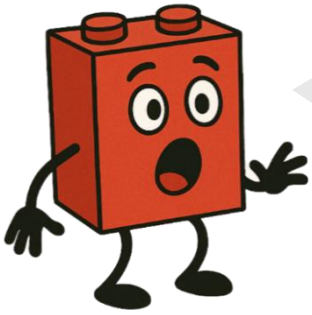
For the blocks, proceed as follows. Divide the steps with your group mates – this will make everything faster! ⏱

- Craft the 8 Bridge Blocks
- Craft the 8 sensor blocks (hot glue gun required)
- Make the note blocks from the leftover blocks. To play well, we recommend 25 large and 8 small blocks. In any case, when editing, keep in mind that **8x 2kOhm** resistors are left for the wiring.
  ***Tip**: If you're short on time, you can skip some notepads.*

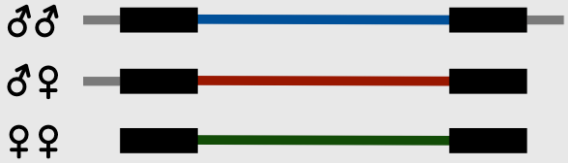








Jumper Cable Types
There are three different types of connectors. They are represented by ♀ and ♂ :



BRIDGE BLOCKS

In order for the note blocks to be connected to the sensor paths, we have to prepare the bridge blocks.

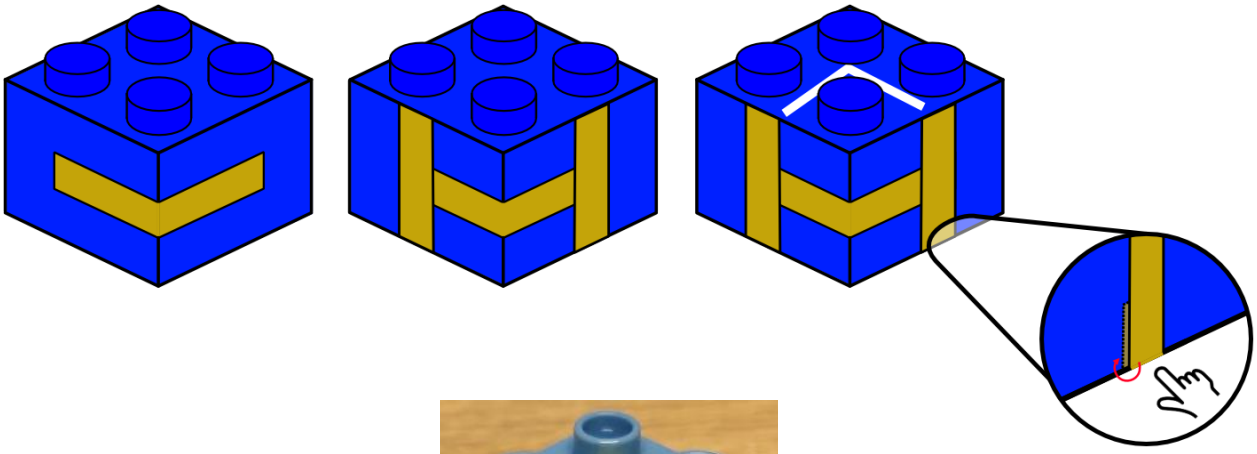
The bridge blocks are designed very simply. They have no resistors, but only use conductive tape to close the circuits.

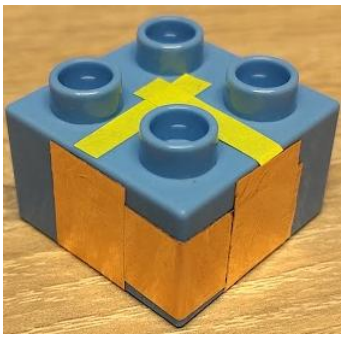
8 x bridge blocks are required. Proceed as follows:

1. Cut one ~4 cm long and two ~2.5 cm long **wide** tapes and remove them from the protective film.

2. Glue the three adhesive strips to the block. Start with the longer, horizontal strip and then glue the two shorter ones.
   Make sure that the conductive adhesive strips go **over the edge** of the block.

3. Glue two small paper adhesive strips to the top of the block to make the bridge block easier to see. (Optional)





This is what it looks like, for example 😊

SENSOR BLOCKS

We need **8 x sensor blocks** that connect the power signal from each sensor track to the Arduino. **Each sensor block is pre-drilled.**

For each sensor block, proceed as follows:

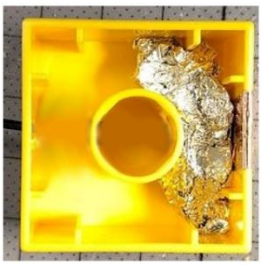
1. Take one of the pre-drilled sensor blocks.



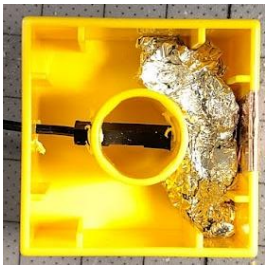
2. Glue a conductive **wide** strip of tape to the **unpierced** side from the inside out. Make sure that the tape inside reaches about the "plastic bottom".



1. Tear off about 20x10 cm aluminum and press it into an aluminum "croissant". Insert this croissant on the side of the adhesive strip in the block. The aluminum must touch the adhesive strip. (Detailed instructions on how aluminum inserts work can be found on page 16)



3. Now insert a ♂♂ **(male-male)** jumper cable with the pin through the hole into the aluminum. **Only use cables with the colors from the illustration in step 5.** Use each color only once. *Tip: With a small ruler or triangle, the pin can be pressed even better into the aluminum.*



4. Label the sensor block at the top with a marker with the correct number. So the sensor block with a red wire says something like "0" or the sensor block with a blue wire "4". Note that **B7 comes in dark brown <u>or</u> purple** .



1. Fix the cable in the block with a hot glue gun at the openings, so that the cable is well stabilized.

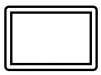




Risk of injury: Use gloves!



Never work unsupervised!



Work on a work surface!

NOTE PADS

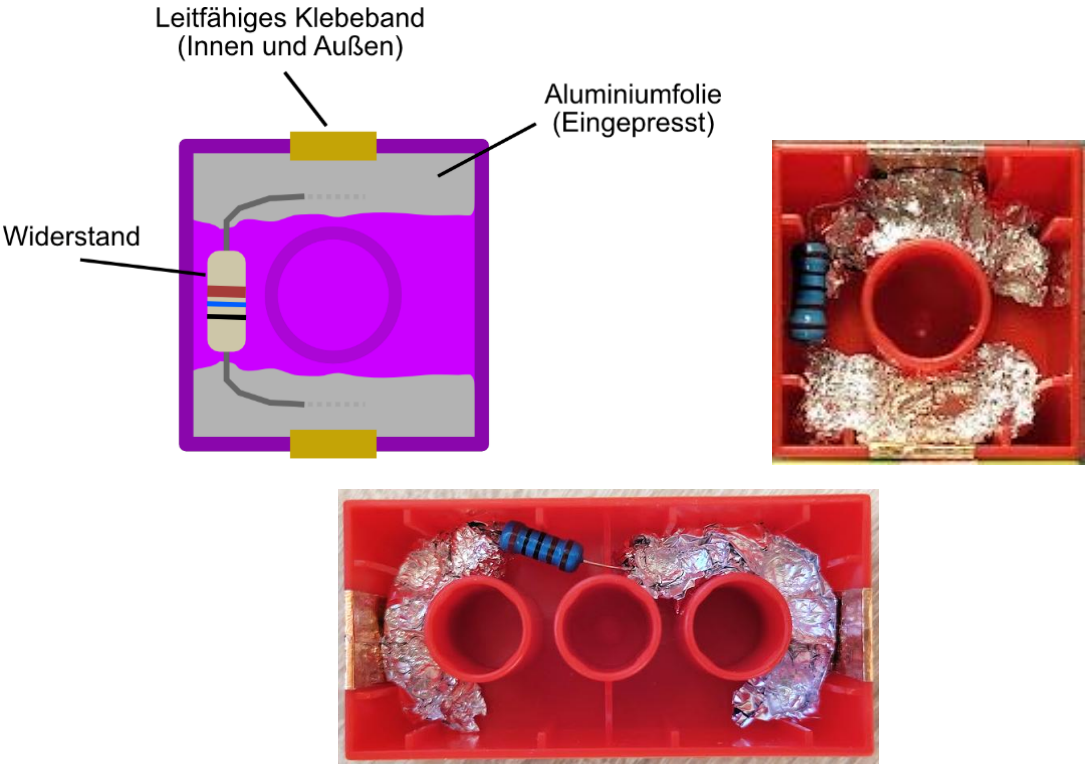
The record is ready: Now the note blocks come into play. We have two different sizes available: 2x2 and 2x4 blocks.

Depending on the size of the block, a different resistor must be installed:

- 2x2 Block: 1kΩ (1000 Ohms)
- 2x4 Block: 2kΩ (2000 Ohms)

The aim is to fix the resistor to the conductive adhesive tape by means of aluminum foil so that the adhesive tape allows a continuous electrical flow.

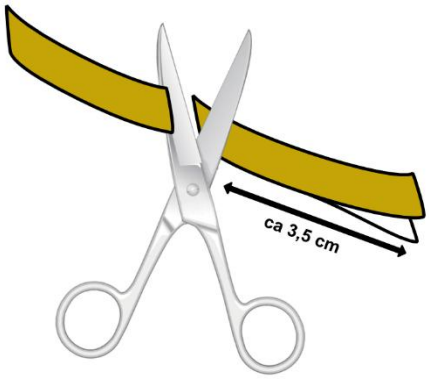
A finished notepad looks like this (view from below):



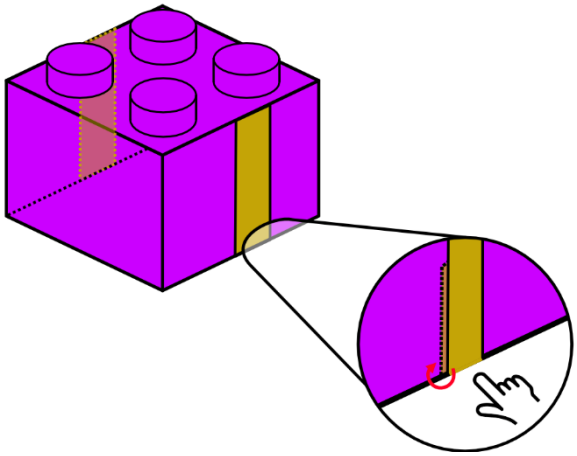
Follow the steps on the following pages in the order given to make a notepad. Repeat the steps until you have finished all (or some) notepads.

The instructions apply to 2x2 and 4x2 blocks. Always choose the right resistance and the right "aluminium croissant size"!

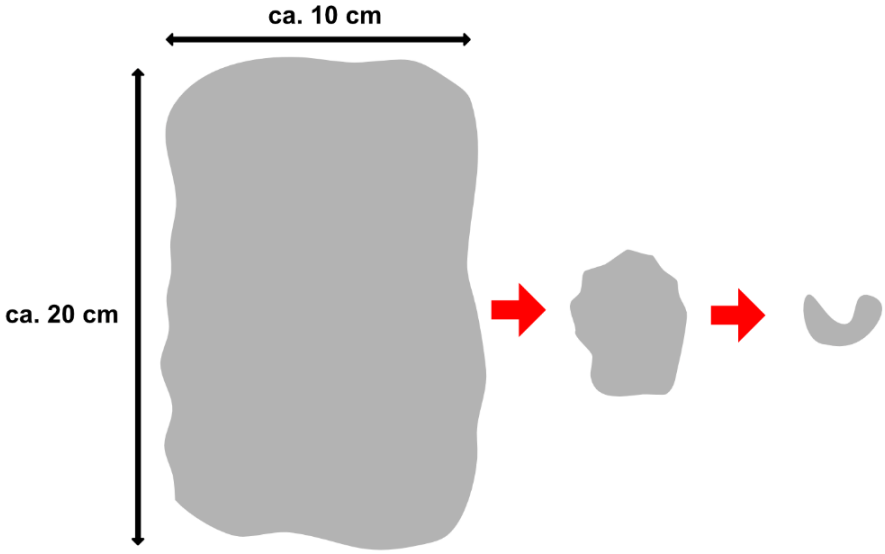
1. Cut **two** adhesive strips about 3.5 cm wide and remove them from the protective film.



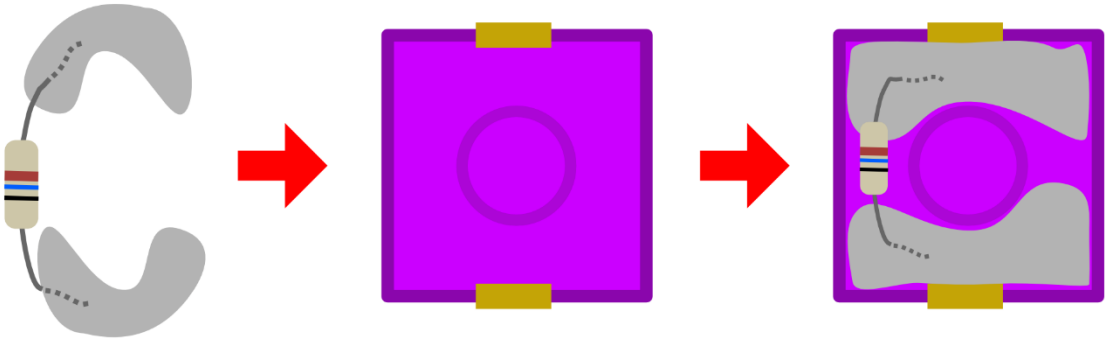
2. Glue two strips of adhesive **(on the opposite side)** from the outside over the edge to the inside. Make sure that the film inside is not too short.



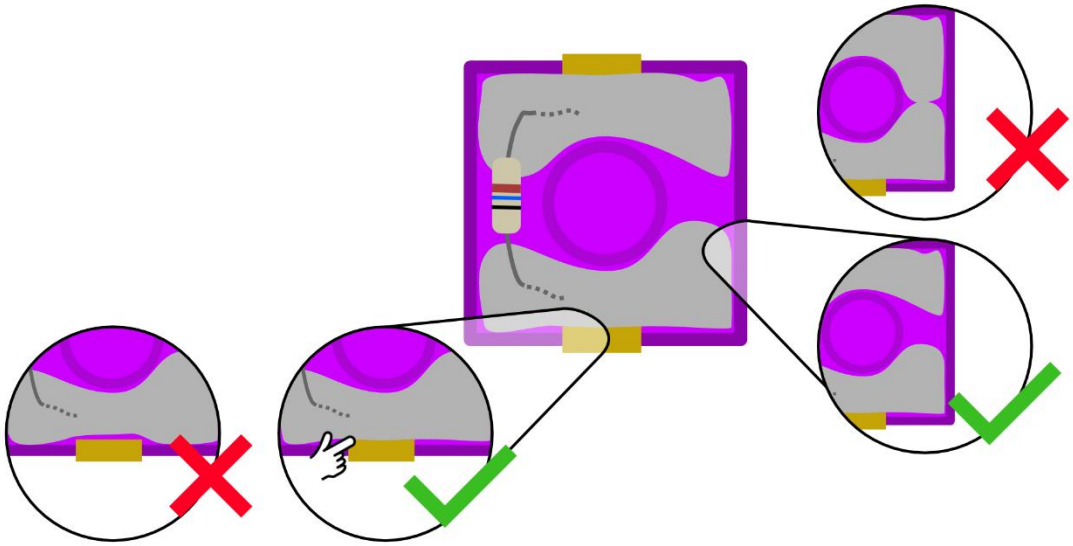
3. Make **two** aluminum "croissants" and shape them so that they fit between the middle plastic circle and the inner wall of the block. Use (***approximately***) 10 x 20 cm aluminum foil.



4. Insert the resistor into both aluminum balls and insert the aluminum resistor construct into the block. You can use a thin pen for this step to press the aluminum crescent into the block.



5. Make sure that the resistor is in place, that the aluminum is touching the conductive tape, and that everything is well pressed into the block. The two aluminum balls must not touch each other[1].



[1] If the aluminium crescents touch, the resistance is skipped – because electricity always takes the shortest path. As a result, the block of notes would not deliver any measurable results 😳

POWER SUPPLY

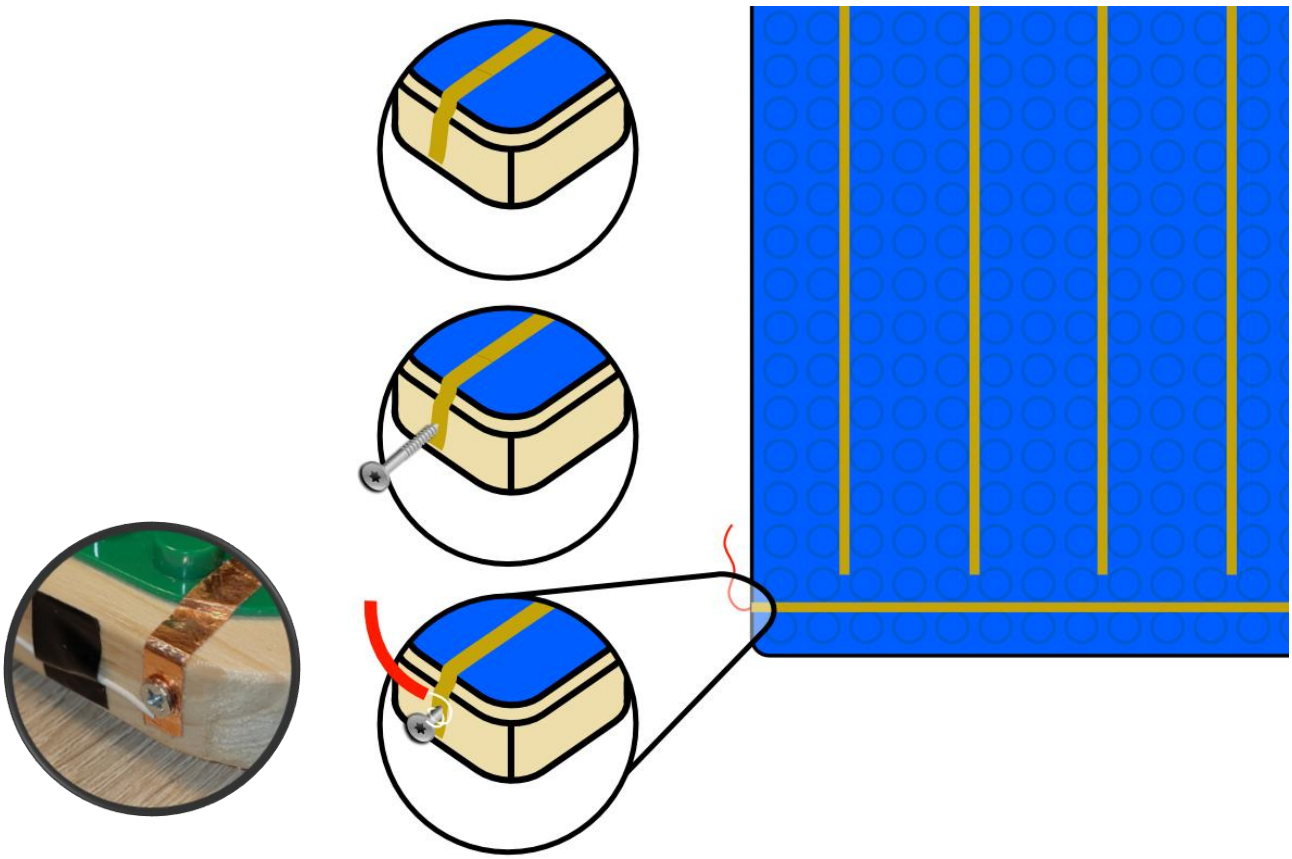
We have already finished building all the necessary blocks. Great!

In the last step, we only have to connect the cable of the power supply to our 5 volt track (running horizontally). We don't need our own block for this, but can attach a cable directly to the narrow conductive adhesive tape.

Perform these steps on one of the pre-drilled holes on the plate. (The hole on the left or right can be used)

Proceed as follows:

- Glue a small piece of conductive **narrow** tape over the edge of the board and connect the 5V track and the hole. The hole is covered by the tape here.
- Screw the enclosed screw into the hole so that there is about 2-3mm of space.
- Now wrap the power cable (stripped end) around the head of the screw. After you have wrapped the cable once or twice, you can screw the screw in the remaining 1-2mm. Make sure that the cable fits well and is in contact with the tape.



ELECTRONICS

After building comes the big wiring. Before we start with the wiring, however, we have to take a quick look at the physical basics.

The Arduino microcontroller has many slots, but we only need a few of them. These are:

- **5V**: This is our power source, which is connected to the 5V train and thus supplies all note blocks with electricity.
- **GND**: That's the zero point. The current flows to this end point.
- **A0 – A7**: These are the individual sensors that are connected to the sensor paths.

Sounds understandable so far. Can we just connect it like that?

No, unfortunately it's not quite that simple. The A0-A7 sensors are **voltage meters**, the resistance is calculated by the voltage between them and GND. Such voltage meters have a very high resistance → So virtually no current flows over the sensor itself.

That's why we have to use a so-called **voltage** divider (eng. Voltage Divider). Such a divider always consists of several resistors. We already have at least one resistor: the note blocks[2]. In addition, we need a **reference resistor.** Every circuit (i.e. for all sensors) needs such a resistor, which has the same ohms for all sensors.

This results in simplified assembly for each sensor:

5V → Notepads (Rx) → Sensor (Ax) Reference → Resistor (R1) → GND

[2] Except for breaks without a bridge block, where there is generally no electricity.

With this composition, we can now calculate the voltage for each sensor:

$$V_{In} = V_{Out} \cdot \frac{R_x}{R_1 + R_x}$$

And with a little reshaping we can calculate the resistance:

$$R_x = R_1 \cdot \frac{V_{Out}}{V_{In} - V_{Out}}$$

With each:

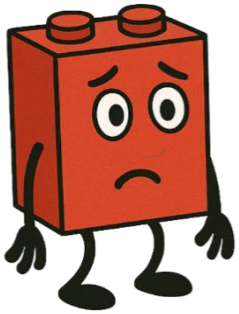
R_x – Resistance through note blocks

R_1 – Fixed reference resistor

V_{Out} – Voltage between Rx and R1 relative to GND

V_{In} – This is the 5V from the Arduino

We use 1kΩ and 2kΩ resistors in our blocks. Accordingly, we also use similarly sized reference resistors. A good trade-off here is a **2kΩ resistor**. We will then add this to the circuit diagram later so that we can finish building our voltage divider.



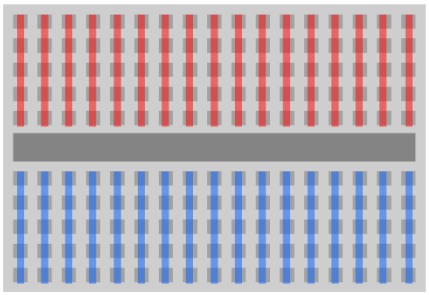
No need to worry!

If you've already desperately unpacked your calculator: Don't worry, we don't have to calculate anything by hand. The AudioBrix software calculates the resistance automatically 😉

WIRING

We wire the previously discussed voltage dividers with a breadboard and jumper cable. Breadboards are empty plates with holes in which jumper cables and resistors can be inserted.

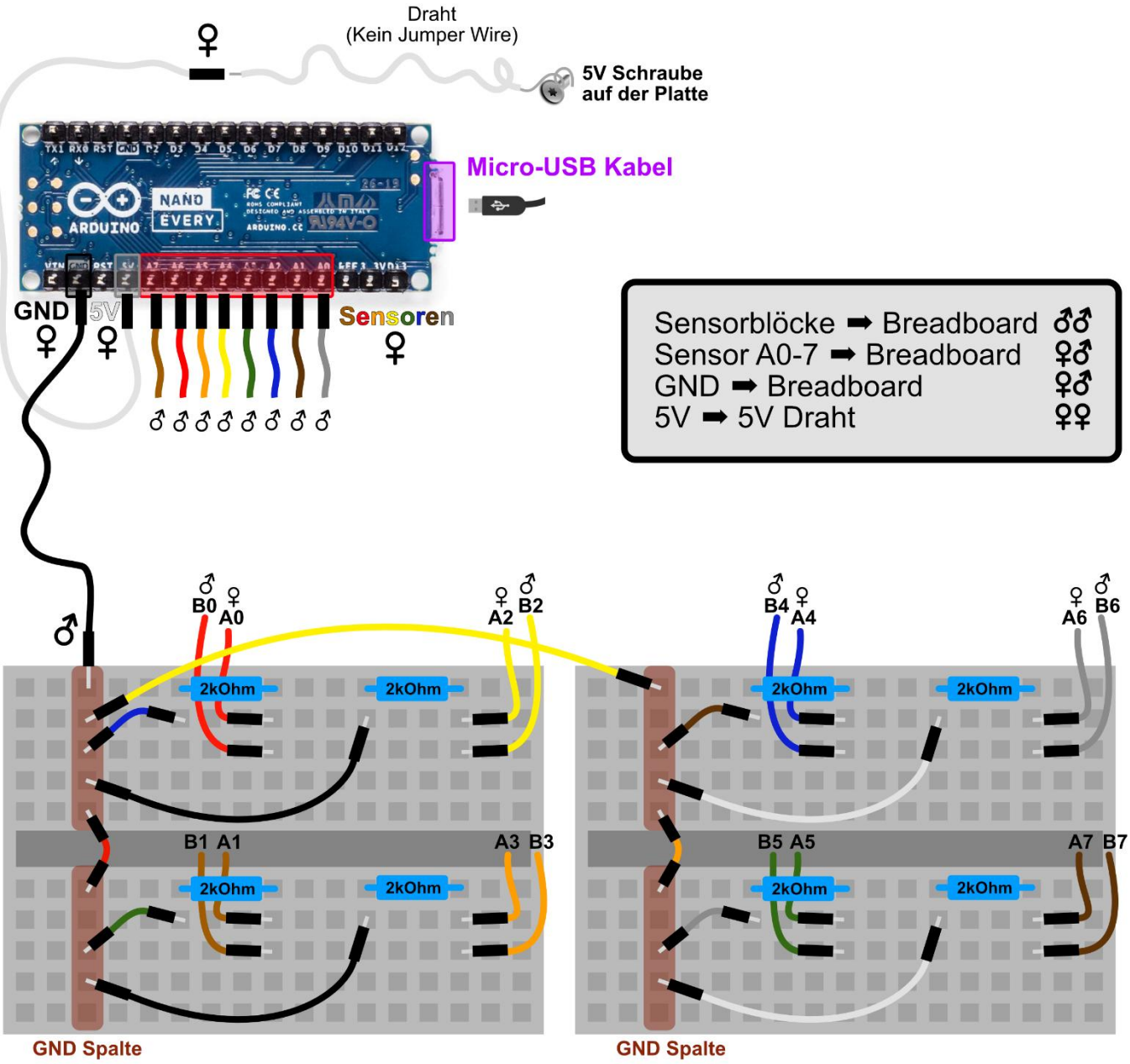
The holes are connected in one direction. In the right image, all the holes within the red and blue lines are connected to each other. In the other direction (transversely) they are not connected. Likewise, the columns are separated with a long bar.



For each music track:

- A reference resistor (2kΩ) is inserted between two columns in the breadboard.
- At one end of the reference resistor, the cable from the sensor block (B...) and the cable to the Arduino sensor (A...) are plugged in.
- At the other end of the reference resistor, a cable to a GND series is placed.
- Use the colors as in the detailed cabling view on the next page.

On the following pages you will find the detailed circuit diagram as well as a detailed step-by-step guide.



When wiring, keep in mind:

- The right colors[3]
- The right cable types (female: ♀ and male: ♂)
- The order of the sensors (A0-A7) and the sensor block cables (B0-B7)
- 2kOhm resistors for the breadboard (shown in blue above)

[3] The 5V cable (♀♀) may vary in color.

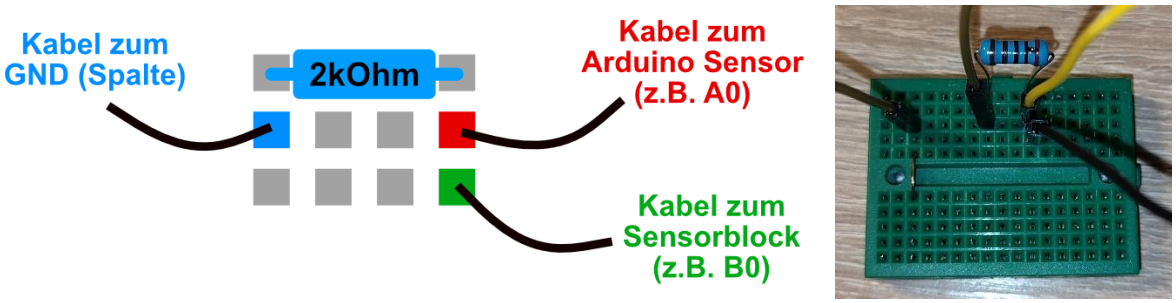
STEP-BY-STEP CABLING

As already mentioned, we need **eight voltage dividers**. Let's take a look at a single voltage divider:

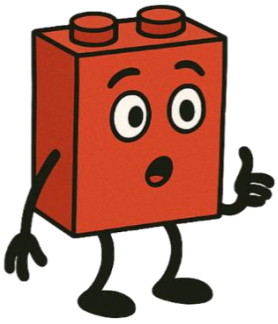
- At one end of the 2kOhm resistor are the cable to the Arduino sensor pin (A0 – A7), as well as a cable to the corresponding sensor block (schematic B0-B7).

  The sensor block (B0) of the first beat belongs to A0, the second beat / sensor block to A1 and so on.

- On the other side of the resistor, a cable goes to a GND column. This GND column connects all the "outputs" of the voltage dividers to the Arduino.



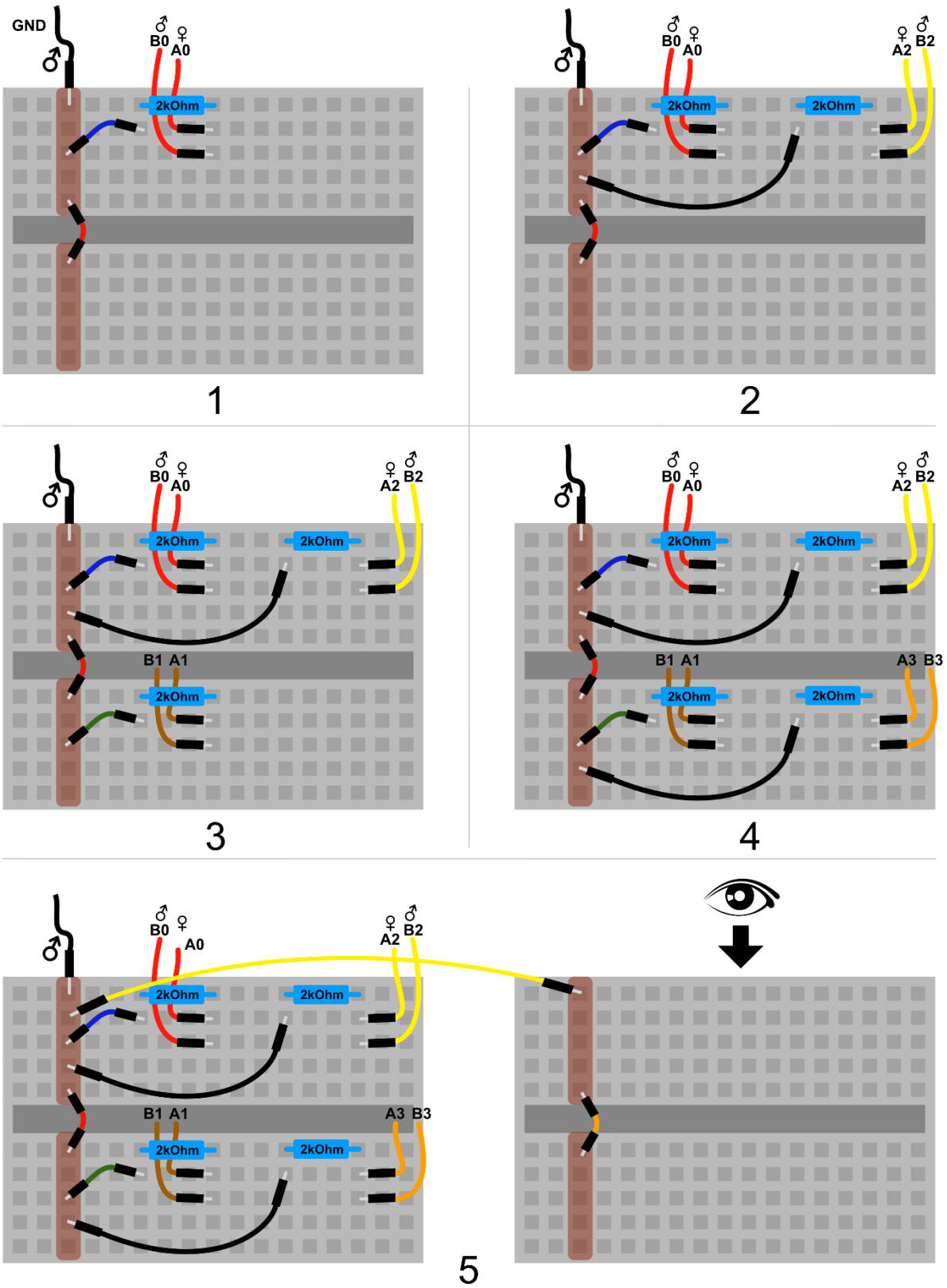
The goal is to place eight voltage dividers on two small breadboards. So four voltage dividers have to be placed per breadboard.



We start by counting from zero!

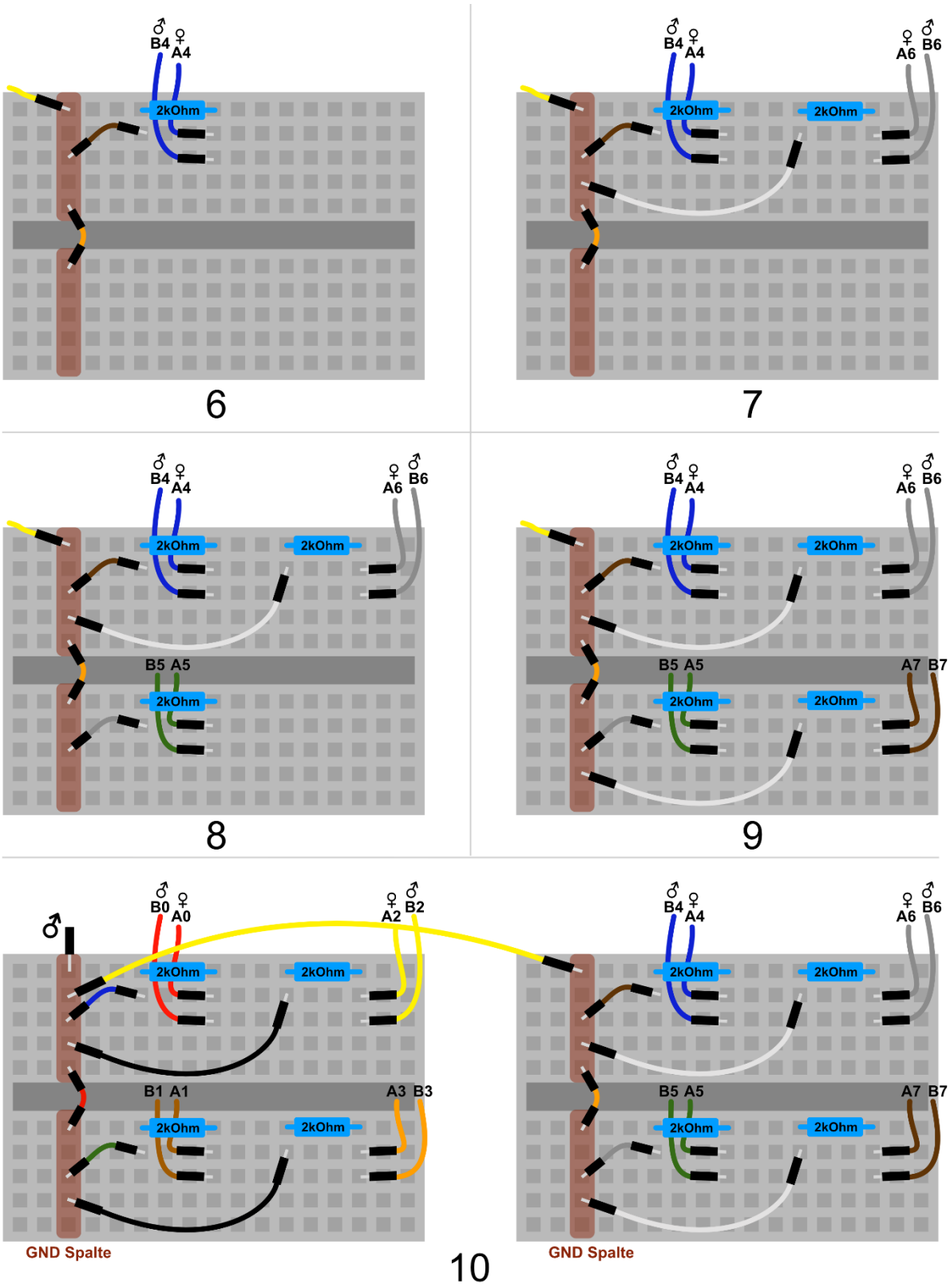
The sensor pins start at 0. Accordingly, we also start at 0 for the sensor blocks.

In computer science, counting has always been from zero since 1967. So if we mean beats 1 to 8, in computer science it means 0 to 7.



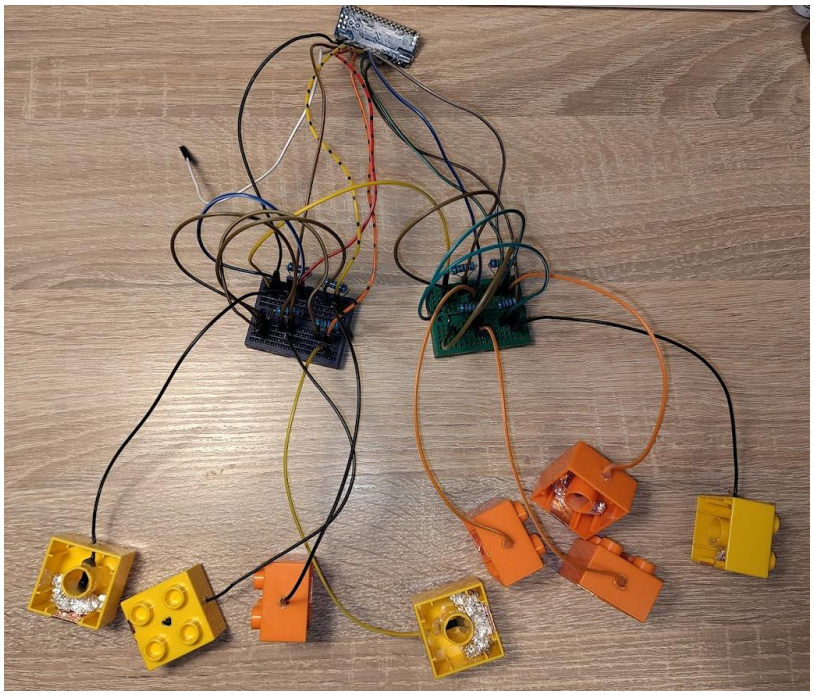
B0-B7 stands for the cables to the sensor blocks
A0-A7 stands for the cables to the Arduino sensor pin

(The exact position of the voltage dividers may differ slightly, but the cables/resistors must always fit accordingly in the column.)

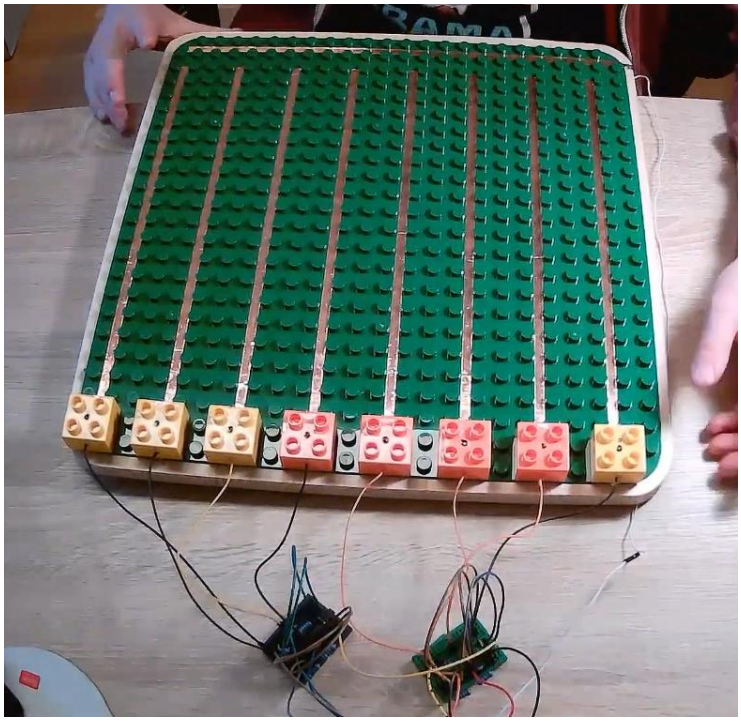


B0-B7 stands for the cables to the sensor blocks
A0-A7 stands for the cables to the Arduino sensor pin

The finished cabling then looks something like this[4]:



Now put the blocks in the correct (color-based) order on the sensor tracks from left (0) to right (7). It looks like this:

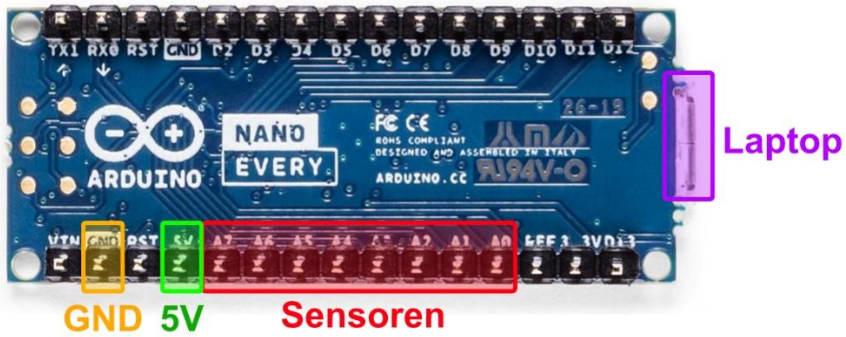


The outputs drawn in the circuit diagram (5V, GND and the sensors) must be plugged into the Arduino. On the back of the Arduino (where the

[4] In this illustration, the color codes deviate from the previously discussed instructions. Instead, the blocks were labeled (B) 0-8. Nevertheless, stick to the color coding as shown in the instructions.

jumper cables are plugged in) are small texts that show where which connection is.

The pins relevant to us are:



After connecting all the necessary cables, the Arduino can be connected to the laptop. Use the micro-USB cable to plug the Arduino into a USB port on the laptop.

Don't forget to connect the power cable of the base plate (which is fixed to the screw) with the 5V cable of the Arduino – otherwise nothing works here!

When the Arduino is connected, the light on the front will flash orange.

SOFTWARE

Our instrument is done – good work! Now all that is missing is the connection via the AudioBrix software. In order for us to be able to use the software on our computer, the following steps must be carried out. There are two variants for this.

Option 1: Open the application using .exe (Recommended)

1. Download the AudioBrix software:- Open dev.waldhoer.org/audiobrix in your browser- Enter the following password: **Brick-Brix-1365**

2. Extract the ZIP folder ***repo.zip*** and open the folder *waldhoer-majorbrix... > dist*.

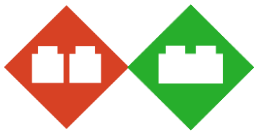
3. Double-click *to open the* main.exe file it contains.

   Caution: It is possible that virus programs may incorrectly mark the application as dangerous. Nevertheless, continue with this warning.

   A terminal will open. Do not close this window.

4. Open your browser and open ***localhost:8000*** in the address bar

If you're having trouble launching the application or encounter error messages in the console, try variant 2 on the following page.



Variant 2: Start the application using Python:

1. Download the AudioBrix software:- Open dev.waldhoer.org/audiobrix in your browser- Enter the following password: Brick-Brix-1365

2. Extract the ZIP folder ***repo.zip*** and open the folder it contains, which contains a *install.bat* or *install.sh* file.

3. Download the Python programming language from the official Python site (www.python.org/downloads) and follow the steps of the installation wizard.

4. Go back to the folder and run the following:- **Windows**: Double-click on *install.bat*
   - **MacOS:** Double-click on *install.shThe necessary libraries will then be installed automatically.*

5. To start the software, navigate to the ***dashboard*** folder and double-click on *main.py*.

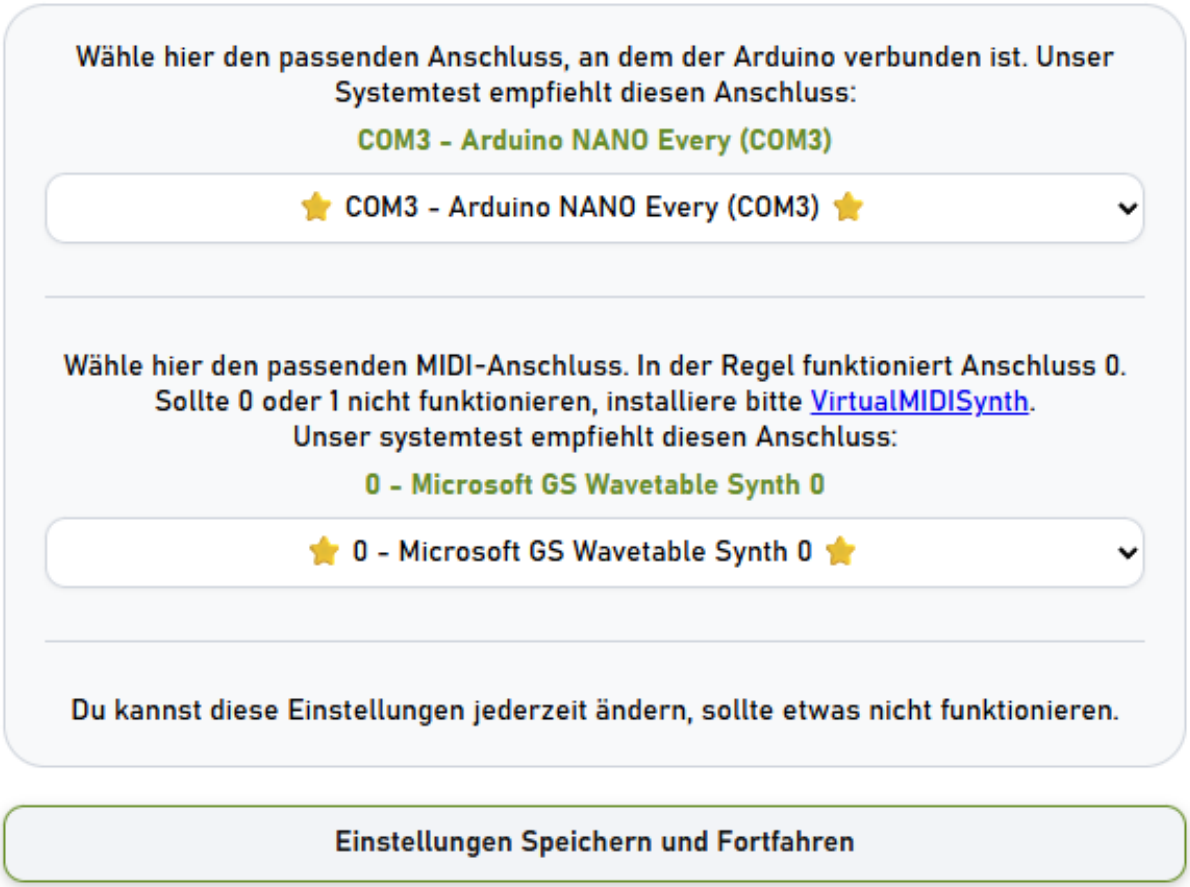
   A terminal will open. Do not close this window.

1. Open your browser and open ***localhost:8000*** in the address bar

SETUP

When starting the application for the first time, small things still have to be set to make the instrument work.

The setup starts automatically when you open it for the first time.

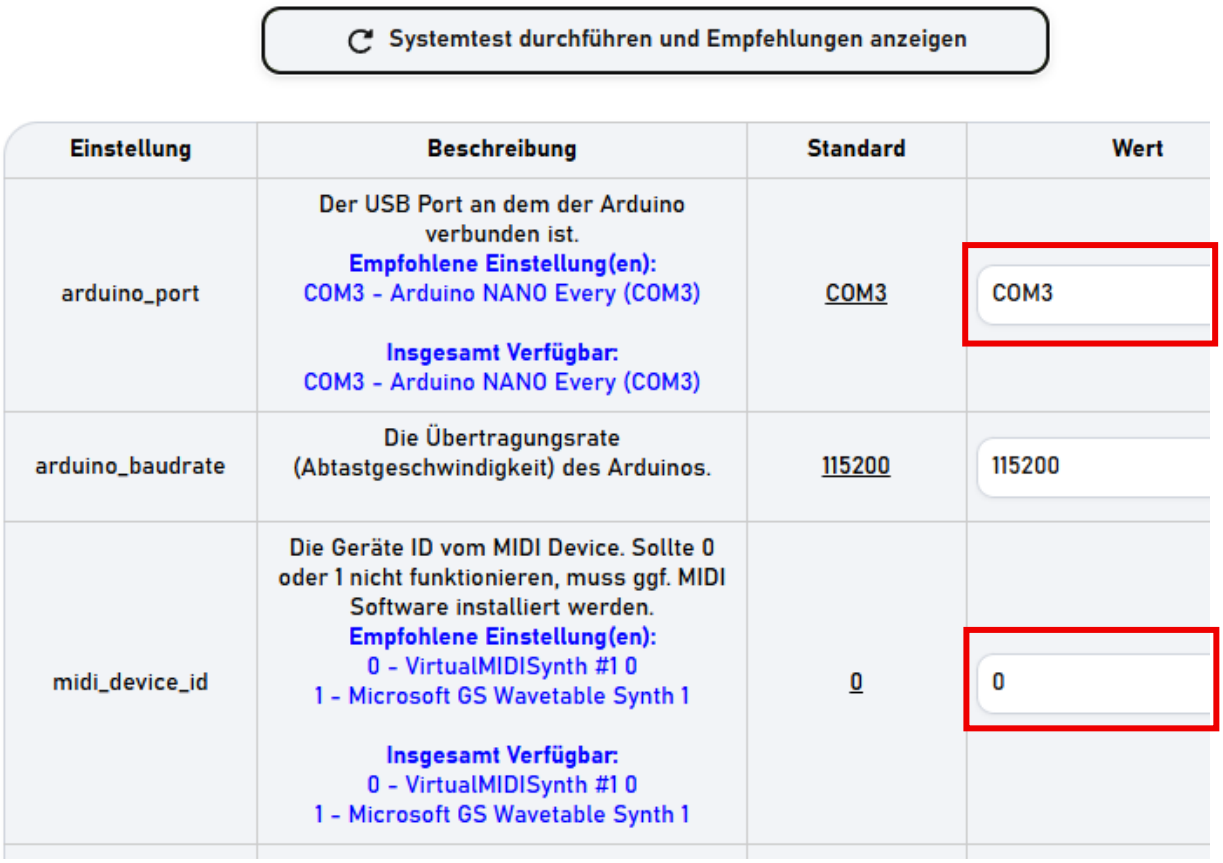


Set these two parameters and click on "Save and Continue". A small tutorial follows and off you go!

In case the sequencer or block detection does not work, we will have to configure the properties ***arduino_port*** and ***midi_device_id*** further. The software will automatically suggest the available options.

1. **Arduino port**: The AudioBrix program usually recognizes the correct connection and shows you the recommendation. Example: "COM3"

- **MIDI Device**: **It is best to try the MIDI synthesizer 0 or the suggested ID first.** If you have problems starting the sequencer later, try another ID. If it still doesn't work, install the free MIDI tool ***VirtualMIDISynth*** (coolsoft.altervista.org/virtualmidisynth). This will then also be displayed in the settings – use this ID accordingly.



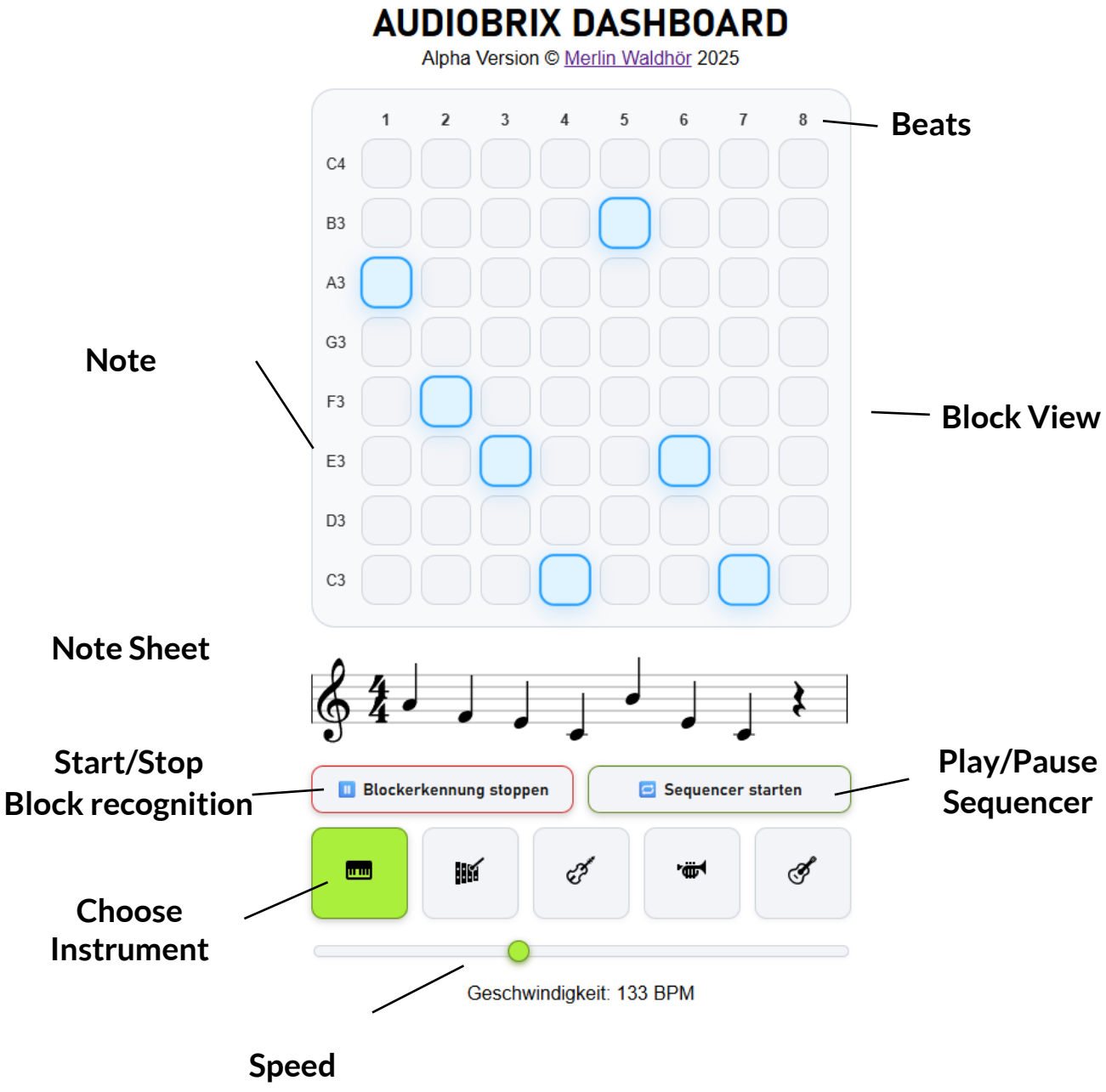
Other settings can also be set up on this page. But beware: Changes can cause errors to be heard!

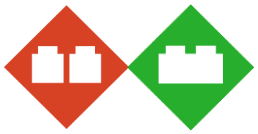
After making the changes, click on "**Save Settings**" and return to the game area.

DASHBOARD

The dashboard is our play area. Here we can

- See and hear the recognized notes
- Start/stop block detection
- Start/Stop the Sequencer
- Changing Musical Instrument
- Switch speed
- View details of the program
- Mute Beats





1. **Block View**:
   The top table-like view shows the currently placed note blocks. Press one of the beats at the top (1 to 8) to mute a beat. Muted columns are shown in red.

2. **Note View**:
   The Note View displays the notes as musical notation.

3. **Block Detection**: When Block Detection is enabled, the resistances are measured and note values are calculated. When Block Detection is turned off, no updates (when → the blocks are replaced) are displayed.

4. **Sequencer:**
   If the sequencer is activated, the notes are played in the loop. If it is deactivated, no notes are played.

5. **Instrument:**
   You can choose between several sounds. To choose a sound, click on one of the symbols (piano, xylophone, strings, trumpet, guitar)

6. **Speed:**
   The playback speed can be adjusted with the slider. The speed is given in BPM ("beats per minute"). It only affects the sequencer.

7. **Detailed output**:
   Errors can occur again and again. In this case, it makes sense to display the detailed log console. Helpful information is often revealed there. Common problems are answered in the chapter "Frequently Asked Questions".

FREQUENTLY ASKED QUESTIONS

SOFTWARE

1. **I see a warning, what does that mean for me?**
   In software in general, a distinction is made between different types of messages. A warning can indicate irregularities that can lead to a problem. However, this does not have to be the case. It is more likely that an error – i.e. an actual problem in the program flow – appears together with a warning. Warnings can help us to recognize the problem.

2. **I've already tried MIDI Device 1 and 0, but the sequencer still won't start.**
   On Windows, it's (unfortunately) not uncommon for the default MIDI provider to not work. In this case, we need an additional MIDI tool to use the instrument.

   VirtualMIDISynth is one such tool that can be installed for free. After installation, the MIDI device will be suggested in blue in the settings. Select the number displayed and try again!

3. **The sequencer is running, but I don't hear anything. Why?**
   This occurs if, for example, VirtualMIDISYnth has been installed, but no SoundFont is activated in it. To do this, open the VirtualMIDISynth settings and click on "SoundFonts". Download any SoundFont from the website: https://coolsoft.altervista.org/en/virtualmidisynth#soundfonts and add it with the "+" button.

4. **The dashboard doesn't work or hangs, what should I do?**
   Sometimes there can be errors in the dashboard. Reload the page in the web browser (usually F5 key) and try again. If the problem persists, try restarting the program by closing the terminal and reopening it using main.exe.

5. **Warnings and errors are sometimes in English. Why?**
   This is because internal program processes and problems are usually displayed in English. We have translated common problems – but unfortunately this is not possible for all errors

6. **I have discovered an error in the program that was not caused by me – who can I report to?**
   The program is in continuous development. Feel free to report your problems by e-mail to merlin@waldhoer.org. Please send the file "events.log" in the "log" folder at the same time, so that a meaningful evaluation can be made.

WIRING

1. **Is working with electricity dangerous?**
   Basically, working with electricity always involves dangers. In our case, however, we can rest assured: The Arduino works with far too little power for the human body to be affected by it - virtually no current flows over the body.

   Nevertheless, working with electrical circuits - for example at sockets at home - is life-threatening and should always be carried out by qualified personnel. Ask the teaching staff if you are unsure about this.

2. **The detected note blocks fluctuate a lot, how can I fix this?**
   Occasionally, there are always minor fluctuations in the resistance values – this is quite normal. However, if a deviating result occurs again and again with certain blocks, it is advisable to stick additional conductive tape on the conductive sides. This may improve the contact with the next block. Also check that the inside of the block is properly seated and that no parts are loose.

   If you are sure that the blocks are well connected and that there are still fluctuations, also check your wiring. With the breadboard, resistors or cables can easily slip out of the socket – this can lead to errors.

3. **My Arduino Nano lights up continuously, not flashing. What to do?**
   In this case, there is probably a defect in the Arduino process. Press the white button on the Arduino briefly to restart the microcontroller.

4. **What happens in the event of a faulty circuit? What are the risks?**
   It is also possible to make mistakes when working with the Arduino. The most problematic case for us is the electronic short circuit, i.e. that current flows from 5V to the GND without resistance. Most of the time, the computer notices that something is wrong with the current flow and disconnects the device automatically and at lightning speed. However, this is not guaranteed. Therefore, always make sure that the wiring is correct to protect Arduino and computer!

5. **My Arduino is getting hot, what should I do?**
   Unplug the device from the computer immediately! Let the device cool down and check your circuit – there is a serious short circuit. Check the circuit and ask your teaching staff for help.

Developer, Author and Illustrator: Merlin Waldhör, BSc

Supervisor: Dipl.-Ing. Dr.techn. Oliver Hödl

Working Group: Cooperative Systems Research Group

This instruction was carried out as part of the Master's internship P1 and P2 in the Master's program in Computer Science.

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